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ABSTRACT

Epitaxial Mn-doped indium tin oxide (ITO) films were deposited on single-crystal yttria stabilized zirconia (YSZ) substrates with (111), (110), and (100) crystal plane orientations using RF magnetron sputtering. The epitaxial relationship between the Mn-doped ITO films and the YSZ substrates was studied using x-ray diffraction (XRD) patterns in the ω–2θ scan mode and XRD pole figures. The Mn-doped ITO films on the YSZ(111) and YSZ(110) substrates exhibited a higher degree of crystallinity than the film on the YSZ(100) substrate as per the x-ray rocking curves. Fluctuations in the crystalline alignment were found to significantly influence the electrical properties of Mn-doped ITO films. Ferromagnetic hysteresis loops were observed at room temperature for all the epitaxial Mn-doped ITO films, irrespective of their crystallographic orientation. The magnetic properties of the epitaxial Mn-doped ITO films suggest that a combination of delocalized charge carrier-mediated interaction and bound magnetic polaron-driven interaction is required to explain the origin of ferromagnetism in these films. The Mn-doped ITO film on the YSZ(111) substrate exhibited the most desirable characteristics in terms of crystallinity, surface smoothness, electrical conductivity, and magnetic properties.

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I. INTRODUCTION

Diluted magnetic semiconductors (DMSs)[1–6] have garnered considerable attention in recent years as one of the most promising materials for applications in spintronic devices.[7] High-temperature ferromagnetic DMSs are especially suitable for fabricating spintronic devices. Room-temperature ferromagnetism has been observed in oxide-based DMSs,[8–44] including transition metal-doped indium tin oxide (ITO).[17–44] ITO films are widely used as transparent conducting films, which are an important component of various devices such as solar cells and display panels. ITO films can be prepared using several methods, such as magnetron sputtering, pulse laser deposition, thermal evaporation, chemical vapor deposition, sol–gel process, and spray pyrolysis. Among these methods, magnetron sputtering is most suitable for depositing commercial ITO films over large areas. Integrating transition metal-doped ITO films into magneto-optoelectronic devices is straightforward since there are several devices using ITO films in the market. Mn-doped ITO films have been widely studied over the past one and half decades.[17–28] Philip *et al.* reported high-temperature ferromagnetism in polycrystalline Mn-doped ITO films grown on sapphire (0001), Si/SiO_2, and Si (100) substrates by reactive thermal evaporation. Their samples exhibited an anomalous Hall effect showing that the charge carriers are spin-polarized, revealing the magnetic interaction between itinerant electrons and localized Mn spins.[17] Nakamura *et al.* obtained ferromagnetic thin films of polycrystalline Mn-doped ITO on glass substrates using RF magnetron sputtering.[18] Their films exhibited low electrical resistivity of the order of $10^{-4}\,\Omega\,\mathrm{cm}$ and high optical transmittance in the visible light region. Amorphous Mn-doped ITO films on polyethylene naphthalate substrates were reported to show room-temperature ferromagnetism, high electrical conductivity, and high visible

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transparency.[22] The coexistence of high-temperature ferromagnetism, high electrical conductivity, and high visible transparency in Mn-doped ITO films is a desirable factor for developing new-generation spintronic devices. Wang *et al.* reported that the dependence of electrical resistivity on temperature for the polycrystalline samples of $In_{1.90-x}Mn_{0.1}Sn_xO_3$ with $x = 0.02$ and 0.06 showed typical semiconducting behavior in a temperature range from 350 down to 5 K.[19] On the other hand, Kumar *et al.* reported that the temperature dependence of the resistivity of polycrystalline Mn-doped ITO films, which was estimated to be 1.6 and 4.3 at % Mn, revealed a metal-like behavior in a temperature range from 20 up to 80 °C.[25] Mn-doped ITO is considered to exhibit a semiconductor to metal transition, depending on the sample preparation conditions such as Sn dopant concentration. Kundu *et al.* reported that the coercivity of the ferromagnetic film of polycrystalline $In_{1.69}Sn_{0.19}Mn_{0.12}O_3$ varied within 75–250 Oe over the temperature range 10–300 K.[23] The local structures surrounding the Mn ions in the polycrystalline Mn-doped ITO films have been investigated by using x-ray absorption spectroscopy.[26] The Fourier transform of the extended x-ray absorption fine structure (EXAFS) spectrum indicated that the Mn ions preferably substitute the In ions at the *b* sites of the In_2O_3 lattice.[26] The valence of the Mn ions was evaluated to range from +2 to +3 on the basis of the threshold energy obtained from the inflection point of the edge in the x-ray absorption near edge structure (XANES) spectrum.[26] The dominant charge state of Mn in the polycrystalline Mn-doped ITO films was also estimated to be +2 using x-ray photoelectron spectroscopic (XPS) measurements.[25] Although the incorporation of Mn^{2+} ions reduces the carrier concentration of the ITO films, doping by Mn^{2+} ions having higher magnetic moments ions is desirable for improving the magnetic properties of the Mn-doped ITO films. Although polycrystalline Mn-doped ITO films have been extensively studied as above mentioned, the origin of high-temperature ferromagnetism in the Mn-doped ITO is not well understood.

Several studies have shown that the electrical performance of polycrystalline ITO films depends significantly on their preferred crystallographic orientation.[45–56] Epitaxial In_2O_3 films deposited on yttria stabilized zirconia (YSZ) substrates have been shown to exhibit a high electron mobility of 110 $cm^2 V^{-1} s^{-1}$.[57] It is considered to be important to understand how the crystallographic orientation improves the physical properties of Mn-doped ITO films. Although polycrystalline Mn-doped ITO samples have been investigated widely, there are very few studies on single-crystalline samples and/or epitaxial films of Mn-doped ITO. In this study, epitaxial Mn-doped ITO films were deposited on YSZ(111), YSZ(110), and YSZ(100) substrates using RF magnetron sputtering to investigate the dependence of the electrical and magnetic properties of these films on the crystallographic orientation. In particular, epitaxial Mn-doped ITO films deposited on a YSZ (111) substrate exhibited a low resistivity of 1.34×10^{-4} Ω cm with a high electron mobility and carrier concentration. High-temperature ferromagnetism was also observed in these films.

II. EXPERIMENTAL SECTION

A. Sample preparation

Epitaxial Mn-doped ITO films were prepared using RF magnetron sputtering on single-crystal YSZ substrates with (111), (110), and (100) crystal plane orientations. A target with an atomic ratio of In:Sn:Mn = 1.75:0.2:0.05 was used. Prior to the film deposition, the YSZ substrates were cleaned using an ultrasonic cleaner with pure water, ethanol, and acetone for 7 min each. First, the chamber was evacuated to a low pressure of approximately 7.0×10^{-4} Pa. Next, pure Ar was introduced into the chamber as the sputtering gas. The Ar gas flow rate was 10 SCCM. No O_2 gas was added. Subsequently, the target was presputtered for 60 min to obtain a clean surface before the film deposition. During the sputtering process, the working pressure of the chamber was maintained at 1 Pa. The RF power, substrate temperature, and deposition time were 120 W, 450 °C, and 30 min, respectively.

B. Material characterization

The film thickness was measured by using stylus profilometry. The atomic composition of the deposited films was determined using energy dispersive x-ray (EDX) analysis. The crystalline structure of the films was characterized using x-ray diffraction (XRD). A Ge(220) channel-cut monochromator/collimator was employed in the incident optics to use the Cu $K\alpha_1$ radiation ($\lambda = 0.154\,06$ nm) for the high-resolution XRD measurements. The surface morphology of the films was observed using an atomic force microscopy (AFM). The transport properties (i.e., resistivity, carrier concentration, and electron mobility) of the films were measured by Hall measurements using van der Pauw configuration. The magnetic properties of the films were determined using a superconducting quantum interference device (SQUID) magnetometer.

III. RESULTS AND DISCUSSIONS

The thickness of the deposited films was determined by stylus profilometer measurements. The thickness of the Mn-doped ITO films ranged from 800 to 1000 nm. The incorporation of Mn atoms into the deposited films was confirmed by EDX measurements. The atomic concentration of Mn in the Mn-doped ITO films was found to be 1%, and no other impurities were present in the films. Figure 1 shows the EDX maps of the peak intensity of Mn $K\alpha_1$ acquired at an electron accelerating voltage of 15 kV from the Mn-doped ITO films deposited on different YSZ substrates. The homogeneous distribution of Mn composition was observed in the Mn-doped ITO films. The most direct method for improving the spatial resolution of EDX analysis is by reducing the electron accelerating voltage.[58–62] This improvement comes about by reducing the volume of the sample that the incoming electrons are interacting with. To reduce the x-ray generation volume to the sub-20 nm level requires accelerating voltages of 2 kV or less.[60] Figure 2 shows the EDX maps of the peak intensity of Mn $L\alpha$ acquired at an acceleration voltage of 1.5 kV from the Mn-doped ITO films deposited on different YSZ substrates. No extrinsic elements and impurity phase were detected by the EDX measurements, although these observations cannot deny the possibility of the cluster whose size is smaller than the spatial resolution of EDX.

XRD measurements were performed to investigate the crystalline structure of the Mn-doped ITO films prepared in this work. Figure 3 shows the XRD patterns in the ω–2θ scan mode of the Mn-doped ITO films deposited on different YSZ substrates using the logarithmic scale. The magnified images of the XRD peaks of

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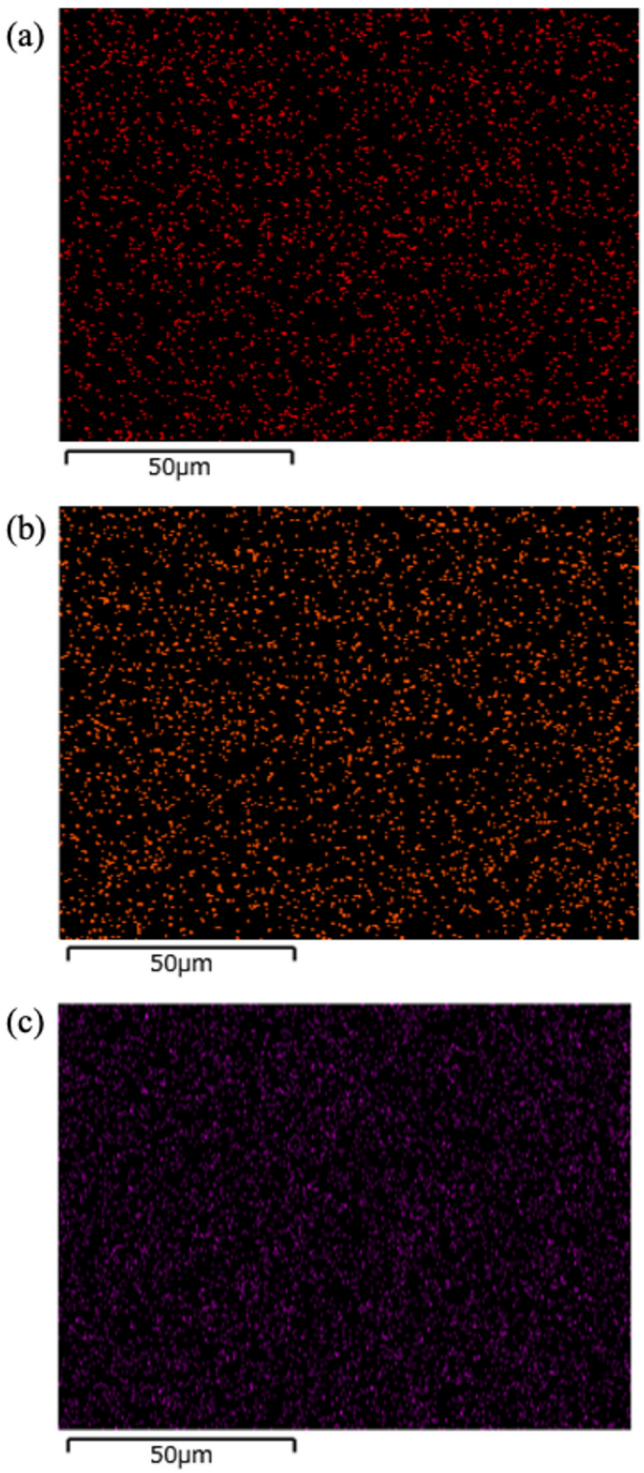


FIG. 1. EDX maps of the peak intensity of Mn $K\alpha_1$ acquired at an electron accelerating voltage of 15 kV from the Mn-doped ITO films deposited on (a) YSZ(111), (b) YSZ(110), and (c) YSZ(100) substrates.

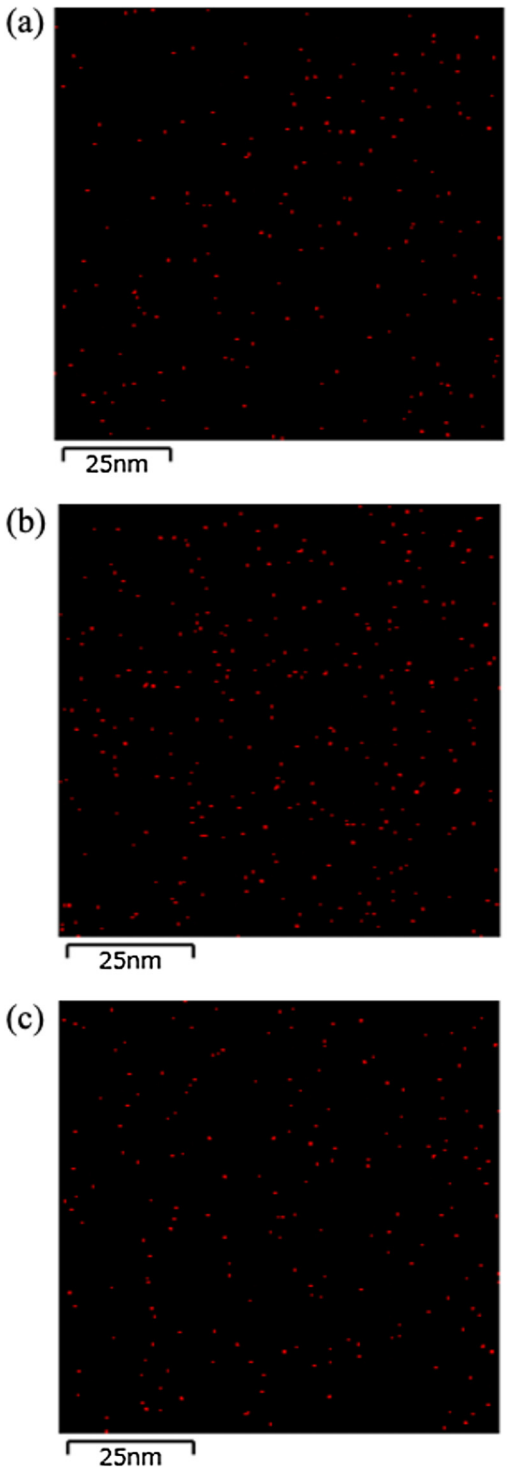


FIG. 2. EDX maps of the peak intensity of Mn $L\alpha$ acquired at an electron accelerating voltage of 1.5 kV from the Mn-doped ITO films deposited on (a) YSZ (111), (b) YSZ(110), and (c) YSZ(100) substrates.

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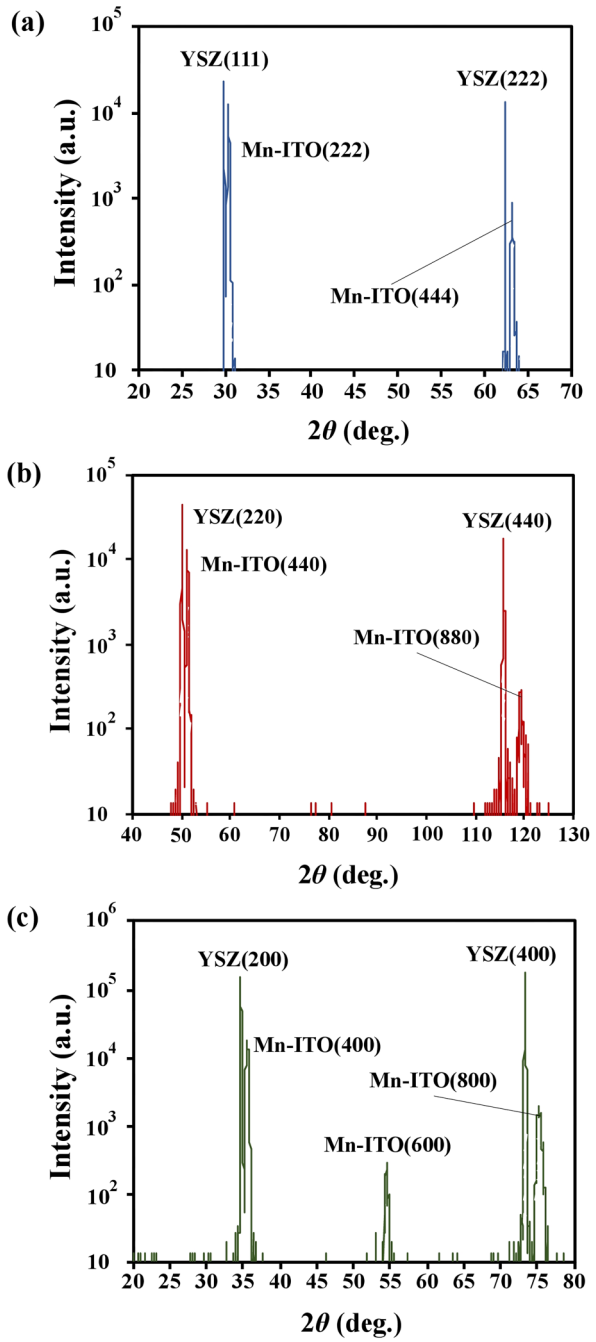


FIG. 3. XRD patterns in the ω–2θ scan mode of the Mn-doped ITO films deposited on (a) YSZ(111), (b) YSZ(110), and (c) YSZ(100) substrates.

the film and substrate over a narrow range of 2θ using linear scale are shown in Fig. 4. The ω–2θ XRD pattern of the Mn-doped ITO film on the YSZ(111) substrate [Fig. 3(a)] exhibited diffraction peaks corresponding to YSZ(111). In addition, diffraction peaks corresponding to the [111] growth direction of Mn-doped ITO were observed, indicating the out-of-plane-oriented growth of Mn-doped ITO on the YSZ(111) substrate. The XRD pattern of the Mn-doped ITO film on the YSZ(110) substrate [Fig. 3(b)] exhibited intense diffraction peaks corresponding to both Mn-doped ITO and YSZ(110). No other diffraction peaks were observed, indicating that [110] is the preferred orientation in this case, which is normal to the surface of the YSZ(110) substrate. The XRD pattern of the Mn-doped ITO film on the YSZ(100) substrate [Fig. 3(c)] exhibited intense peaks diffracted from the (400), (600), and (800) planes of the film as well as those corresponding to YSZ(100). These XRD patterns confirm the out-of-plane-oriented growth of Mn-doped ITO on YSZ substrates, namely, Mn-ITO(111)||YSZ (111); Mn-ITO(110)||YSZ(110); and Mn-ITO(100)||YSZ(100). The lattice constant of the Mn-doped ITO films was calculated to be $a_{\text{Mn-ITO}} = 1.011$ nm from the observed ω–2θ XRD patterns, whereas YSZ has a cubic structure with a lattice constant of $a_{\text{YSZ}} = 0.513$ nm. Thus, $a_{\text{Mn-ITO}}$ is almost two times larger than a_{YSZ}, leading to a small lattice mismatch of 1.4% between the Mn-doped ITO films and the YSZ substrate.

Figure 5 shows the XRD pole figures of the Mn-doped ITO films on different YSZ substrates. The threefold rotational symmetry exhibited by the (400) pole figure in Fig. 5(a) confirms the heteroepitaxial growth of Mn-doped ITO on the YSZ(111) plane. The (440) pole figure of the Mn-doped ITO film on the YSZ(110) substrate [Fig. 5(b)] exhibited one spot at the center, four equivalent spots at a tilt angle of 45°, and two other equivalent spots at a tilt angle of 90°, which confirms the heteroepitaxial growth of Mn-doped ITO on the YSZ(110) plane. Finally, the fourfold rotational symmetry exhibited by the (222) pole figure in Fig. 5(c) corroborates the heteroepitaxial growth of Mn-doped ITO on the YSZ (100) plane.

Figure 6 shows the x-ray rocking curves of the epitaxial Mn-doped ITO films on different YSZ substrates. The x-ray rocking curves corresponding to the (444) and (440) diffraction peaks of the Mn-doped ITO films deposited on the YSZ(111) and YSZ(110) substrates are shown in Figs. 6(a) and 6(b), respectively. The full widths at half maximum (FWHMs) of the (444) and (440) diffraction peaks were 0.28° and 0.24°, respectively, indicating a high degree of crystallinity in these films. By contrast, the FWHM of the (400) diffraction peak in the XRD rocking curve of the Mn-doped ITO film on the YSZ(100) substrate [Fig. 6(c)] was 0.69°, which is significantly larger than those of the films on the YSZ(111) and YSZ(110) substrates. Thus, the YSZ(111) and YSZ (110) substrates yielded epitaxial Mn-doped ITO films with a higher degree of crystallinity.

Several studies have deposited In_2O_3(100) films on a YSZ(100) substrate by adopting a cube-on-cube orientation between the film and the substrate.[63,64] These studies observed a nominal tensile mismatch of 1.6%, which was accommodated by misfit dislocations at the interface. In addition, these studies found that the atomic planes of the epitaxial In_2O_3 islands were tilted relative to the four different in-plane ⟨110⟩ directions of the YSZ substrate.[63,64] Moreover, the surface energies of the (111) and (110) planes were found to be approximately half of that of the (100) planes of the In_2O_3 films.[64,65] Furthermore, submicrometer roughness due to faceting and islanding on length scales greater than a micrometer

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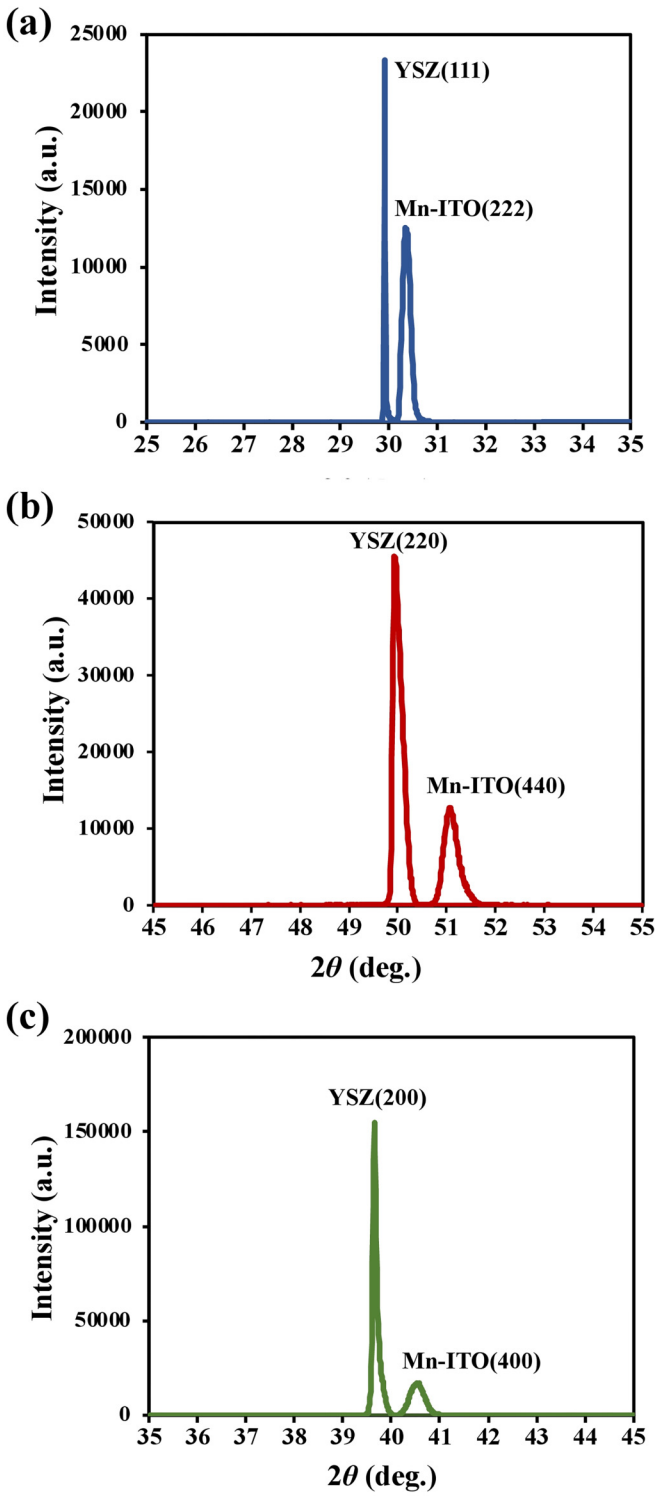


FIG. 4. Magnified images of the XRD peak of the Mn-doped ITO films deposited on (a) YSZ(111), (b) YSZ(110), and (c) YSZ(100) substrates.

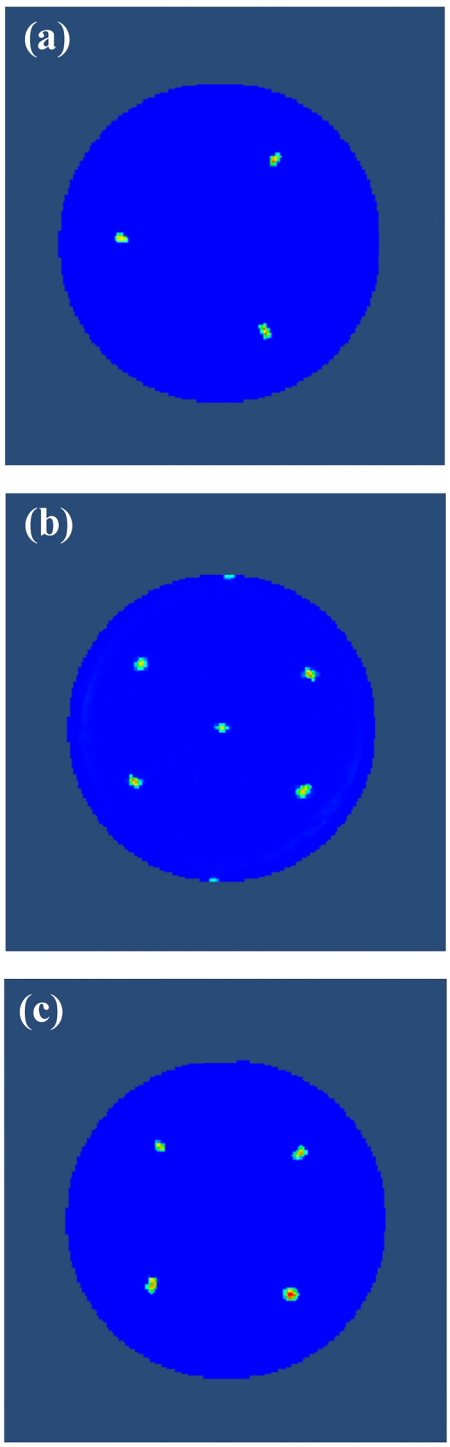


FIG. 5. XRD pole figures of the Mn-doped ITO films corresponding to (a) (400) diffraction of the film deposited on a YSZ(111) substrate; (b) (440) diffraction of the film deposited on a YSZ(110) substrate; and (c) (222) diffraction of the film deposited on a YSZ(100) substrate.

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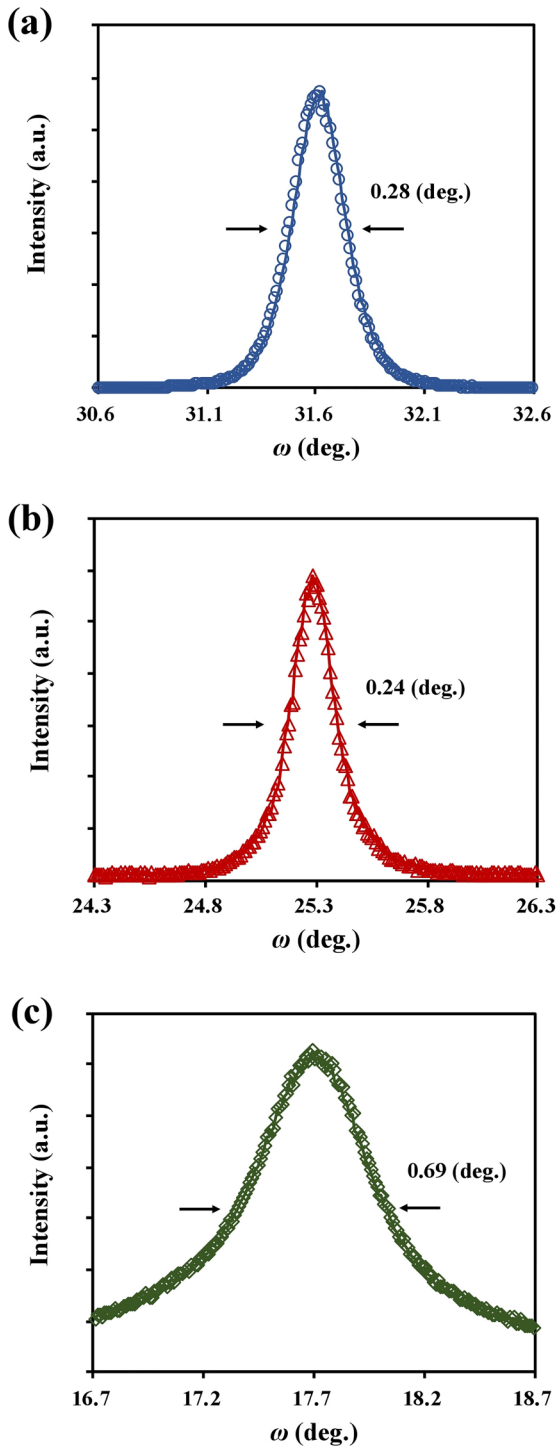


FIG. 6. X-ray rocking curves (ω scan) of the epitaxial Mn-doped ITO films corresponding to (a) (444) diffraction peak of the film deposited on a YSZ(111) substrate; (b) (440) diffraction peak of the film deposited on a YSZ(110) substrate; and (c) (400) diffraction peak of the film deposited on a YSZ(100) substrate.

was reported in nanometer-thick In_2O_3 films epitaxially grown on YSZ(100) substrates via plasma-assisted molecular beam epitaxy.[66] This roughness is believed to arise due to the incomplete wetting of the YSZ(100) substrate by the In_2O_3(100) film and not due to the tensile stress induced by the lattice mismatch between the In_2O_3(100) film and the YSZ substrate.[66] In contrast, In_2O_3 films on YSZ(111) substrates have been shown to grow smoothly without any faceting despite the same lattice mismatch as in the case of the YSZ(100) substrate.[66] Moreover, the surface tension of the In_2O_3(100) films was experimentally confirmed to be significantly higher than that of the In_2O_3(111) films.[67] These findings imply that it is crucial to optimize both the substrate material to ensure the smooth epitaxial growth of Mn-doped ITO films. The adoption of YSZ(111) and YSZ(110) substrates is suitable for the epitaxial growth of Mn-doped ITO films.

The surface morphology of the epitaxial Mn-doped ITO films synthesized in this work was observed using an AFM. Figure 7 shows the AFM images of the epitaxial Mn-doped ITO films on different YSZ substrates. The surface roughness did not exhibit any significant dependence on the crystallographic orientation of the film. The average roughness of the Mn-doped ITO films deposited on different YSZ substrates was found to be approximately 0.3 nm. Such smooth epitaxial films can be used as an intermediate layer for various device applications.

The electrical transport measurements of the deposited films were carried out at room temperature. The various transport properties of the epitaxial Mn-doped ITO films were evaluated by performing Hall measurements in the van der Pauw configuration, which are summarized in Table I. The resistivity of the Mn-doped ITO films ranged from 1.34×10^{-4} to 2.68×10^{-3} Ω cm. In particular, the film on the YSZ(111) substrate exhibited the lowest resistivity. In general, the resistivity depends on the electron mobility and carrier concentration. The resistivity of undoped epitaxial ITO films on YSZ substrates has been reported to be in the range 10^{-4}–10^{-5} Ω cm, which corresponds to an electron mobility in the range 10–42 $cm^2 V^{-1} s^{-1}$ and carrier concentration in the range of 1.74×10^{20}–2.15×10^{21} cm^{-3}.[68–76] Similar to the electron scattering mechanism in undoped ITO films, the electron mobility in Mn-doped ITO films is mainly affected by neutral impurities, ionized impurities, and grain boundaries at room temperature. Although the epitaxial Mn-doped ITO film on the YSZ(100) substrate exhibited a relatively low electron mobility of 4.50 $cm^2 V^{-1} s^{-1}$, the Mn-doped ITO films on the YSZ(111) and YSZ(110) substrates exhibited a high electron mobility of 38.37 and 18.65 $cm^2 V^{-1} s^{-1}$, respectively. In addition, the (111)-oriented and (110)-oriented Mn-doped ITO films exhibited a higher degree of crystallinity than the (100)-oriented Mn-doped ITO films based on the FWHM of the out-of-plane x-ray rocking curves. Such a highly crystalline In_2O_3 host lattice is likely to have a low density of neutral scattering centers and high Sn-doping efficiency, which, in turn, implies that only a small number of dopant atoms would be trapped in the crystal defects and a large number of electrically active species would exist inside the grains.[69] The electron mobility of polycrystalline ITO films consisting of nanometer-sized grains has been reported to increase with increasing grain size (i.e., for grain size comparable to the electron mean free path).[77] This increase in electron mobility is due to grain boundary scattering in

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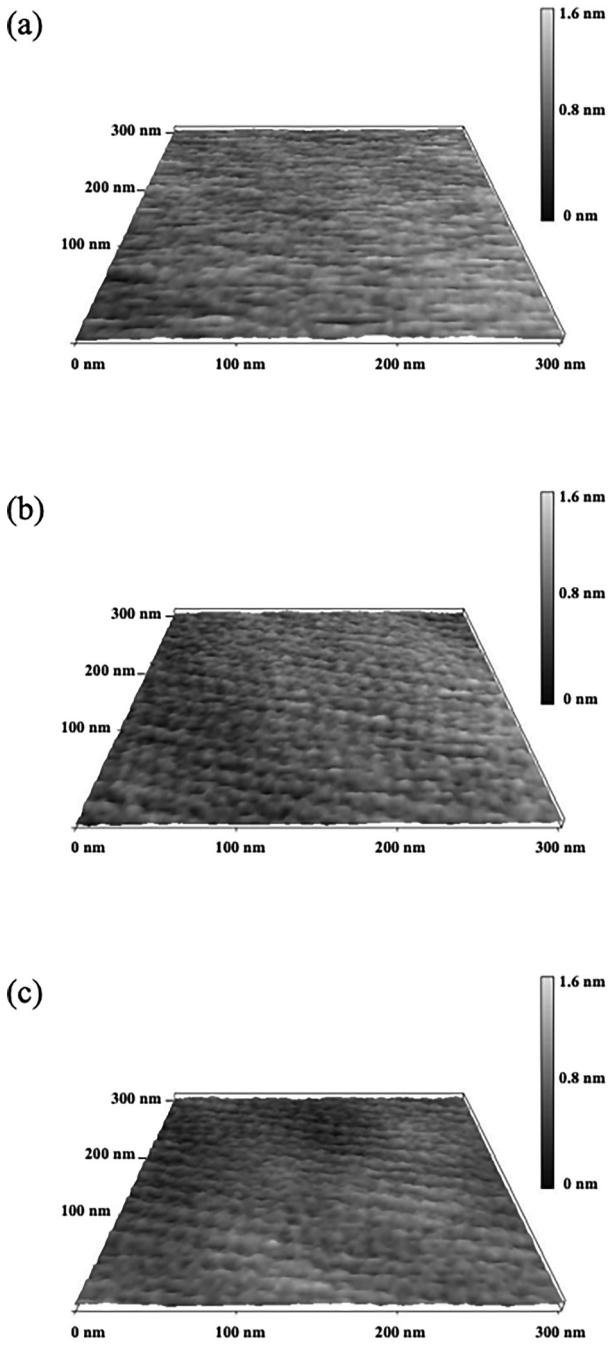


FIG. 7. AFM images of the epitaxial Mn-doped ITO films deposited on (a) YSZ (111), (b) YSZ(110), and (c) YSZ(100) substrates.

polycrystalline films.[77,78] Antiphase domains have been observed using transmission electron microscopy in epitaxial In_2O_3 films deposited on YSZ(100) substrates.[79] Antiphase boundaries can act as barriers for carrier transport in such epitaxial In_2O_3 films.[79]

TABLE I. Transport properties of epitaxial Mn-doped ITO films deposited on YSZ substrates with different crystal plane orientations.

Substrate	Resistivity (Ω cm)	Electron mobility ($cm^2 V^{-1} s^{-1}$)	Carrier concentration (cm^{-3})
YSZ(111)	1.34×10^{-4}	38.37	1.21×10^{21}
YSZ(110)	2.54×10^{-4}	18.65	1.32×10^{21}
YSZ(100)	2.68×10^{-3}	4.50	5.18×10^{20}

The crystal growth orientation has a significant effect on the crystallographic shear plane (CSP)-based grain boundary scattering observed in epitaxial Ta-doped SnO_2 films on TiO_2 substrates having different crystal plane orientations.[80] Antiphase boundary scattering as well as CSP-based grain boundary scattering are also believed to significantly influence the electron mobility of epitaxial Mn-doped ITO films deposited on YSZ substrates.

The carrier concentrations of the epitaxial Mn-doped ITO films on the YSZ(111) and YSZ(110) substrates were similar; however, the carrier concentration of the film on the YSZ(100) substrate exhibited a significantly lower value. The carrier concentration of Mn-doped ITO films is mainly affected by the number of Sn ions, Mn ions, and O vacancies. During the Sn doping of In_2O_3, the In^{3+} sites in the bixbyite In_2O_3 lattice are occupied by Sn^{4+} ions, leading to the generation of free carriers in the ITO films;[81] thus, Sn doping creates not only donor centers but also ionized scattering centers.[82–87] However, increasing the Sn concentration leads to the formation of defect complexes such as electrically inactive Sn atoms coordinated by seven or eight O anions.[82–87] These deactivated Sn atoms act as neutral scattering centers that do not contribute to the carrier concentration of the ITO film.[82–87] The relatively low carrier concentration and electron mobility of the epitaxial Mn-doped ITO film on the YSZ(100) substrate can be explained by the formation of a large number of deactivated Sn atoms, which introduces fluctuations in the crystalline alignment. In our previous work, based on x-ray absorption spectroscopic measurements, we proposed that Mn ions with a valence of +2 and +3 prefer the *b* sites of bixbyite In_2O_3 (in polycrystalline Mn-doped ITO films) for substituting the In^{3+} ions.[26] In case of Mn doping, the extra electrons introduced by the Sn dopant and O vacancies enter the *d*-shell of the Mn atoms, such that the Mn^{3+} ions are converted to Mn^{2+} ions.[27,88] Thus, the incorporation of Mn^{2+} ions compensates for the Sn donor atoms, thus decreasing the overall carrier concentration of the Mn-doped ITO films.

Figure 8 shows the magnetization as a function of applied magnetic field at 300 K for the epitaxial Mn-doped ITO films on YSZ substrates with different crystal plane orientations. The insets in Fig. 8 show magnified images of the magnetization behavior in the low magnetic field region. The diamagnetic contribution from the substrate has been subtracted from the raw data. The magnetic field was applied perpendicular to the film plane. A hysteresis loop was observed at 300 K in all the films. The observed coercivity and remanence indicate the existence of ferromagnetism. The present experimental results suggest that the Curie temperature of the epitaxial Mn-doped ITO films is higher than 300 K.

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Figures 9(a) and 9(b) show the saturation magnetization and coercive force of the epitaxial Mn-doped ITO films on YSZ substrates with different crystal plane orientations, respectively. The saturation magnetization has little dependence on the crystallographic orientation of the films, whereas the coercive force of the (100)-oriented film is slightly smaller than that of the (111)-oriented and (110)-oriented films. Delocalized charge carrier-mediated ferromagnetic interaction is one of the most probable mechanisms for explaining the origin of ferromagnetism in oxide-based DMSs with a high carrier concentration.[89,90] Such an interaction occurs between localized Mn spins, which is mediated by delocalized charge carriers.[91–97] As stated above, the carrier concentration of epitaxial Mn-doped ITO films depends on the crystallographic orientation of the films. Although the difference in coercive force between the epitaxial Mn-doped ITO films on YSZ (111), YSZ(110), and YSZ(100) substrates is small, this difference may be attributed to the delocalized charge carrier-mediated ferromagnetic interaction. On the other hand, it seems difficult to explain similarly the carrier concentration dependence of the saturation magnetization. The possibility of the nonmonotonic behavior of the magnetization and/or Curie temperature as a function of the carrier concentration has been theoretically pointed out.[98–100] This nonmonotonic relationship was experimentally found in some oxide DMSs.[101–103] As for the Cr-doped In_2O_3 films, the increase of the electron to Cr ion density ratio has been reported to decrease the magnetization.[101] When the electron to Cr ion density ratio increases, the Fermi level shifts upward and is farther from the e_g band of Cr ions in Cr-doped In_2O_3 films.[101] It is possible that the upward shift of the Fermi level caused by the increase of the carrier concentration weakens similarly the interaction between the magnetic moments of Mn ions and free carriers in the Mn-doped ITO films, which leads to little dependence of the saturation magnetization on the carrier concentration. The x-ray photoemission spectroscopic measurements for clarification of the electronic structure of the Mn-doped ITO films are under way.

Figure 10 shows the temperature dependence of the magnetization of epitaxial Mn-doped ITO films on YSZ substrates in the range of 2–400 K, which was measured using SQUID. The magnetization data were obtained under an applied magnetic field of 1000 Oe perpendicular to the film plane. There was no significant difference in the behavior of the magnetization curves between the Mn-doped ITO films on the YSZ(111), YSZ(110), and YSZ(100) substrates. No indication of any magnetic transition was observed up to 400 K. The concave shape of the curves shown in Fig. 10 can be explained by using the bound magnetic polaron (BMP) percolation model.[104] In this work, the film deposition under the atmosphere of pure Ar gas promoted the generation of O vacancies in the Mn-doped ITO films. The percolation of BMPs is mediated by shallow donor electrons trapped in O vacancies, which generates a long-range ferromagnetic exchange interaction.[104–106] This implies that not only delocalized charge carrier-mediated interaction but also BMP-driven interaction is necessary to explain the origin of ferromagnetism in epitaxial Mn-doped ITO films.[90,102,107] However, it is too early to draw a conclusion on the mechanism of the ferromagnetism in magnetic transparent conducting films. More experimental information on the carrier-mediated exchange interaction is needed to make a detailed argument on the

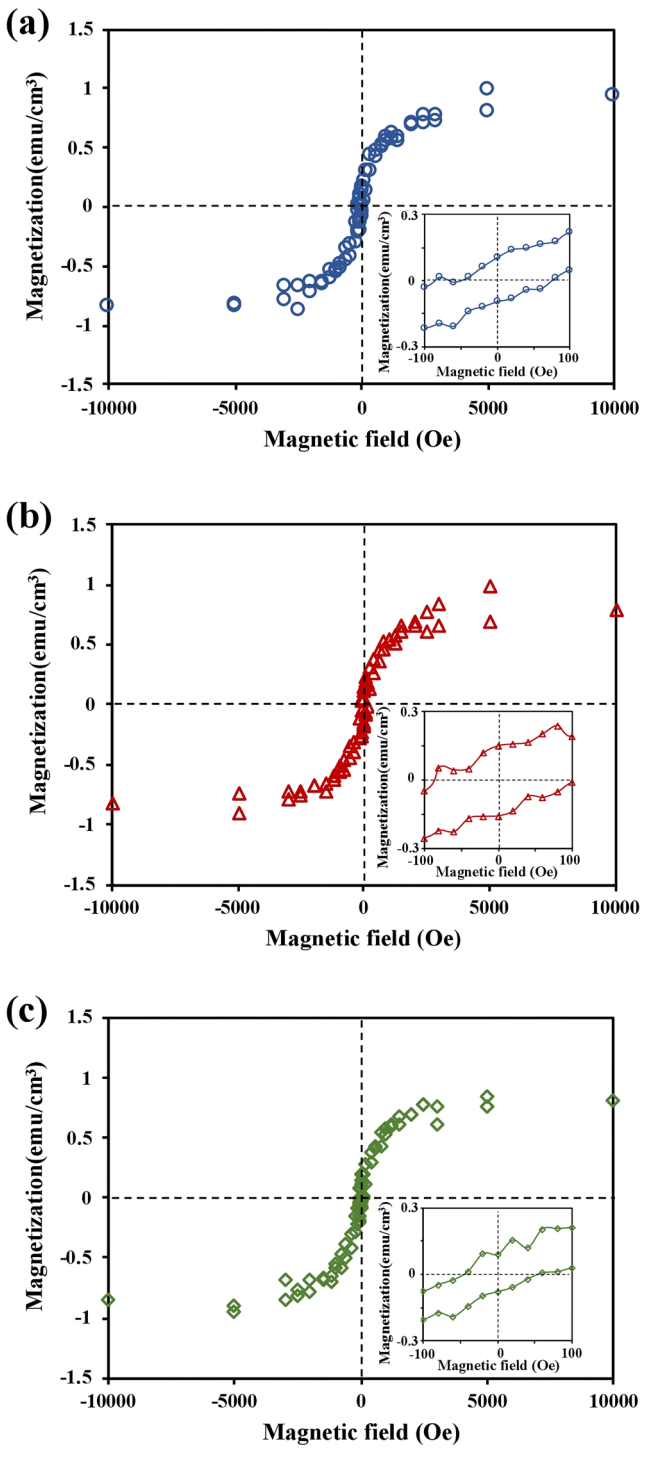


FIG. 8. Magnetization as a function of applied magnetic field at 300 K for the epitaxial Mn-doped ITO films on (a) YSZ(111), (b) YSZ(110), and (c) YSZ(100) substrates. The insets show magnified images of the magnetization behavior in the low magnetic field region.

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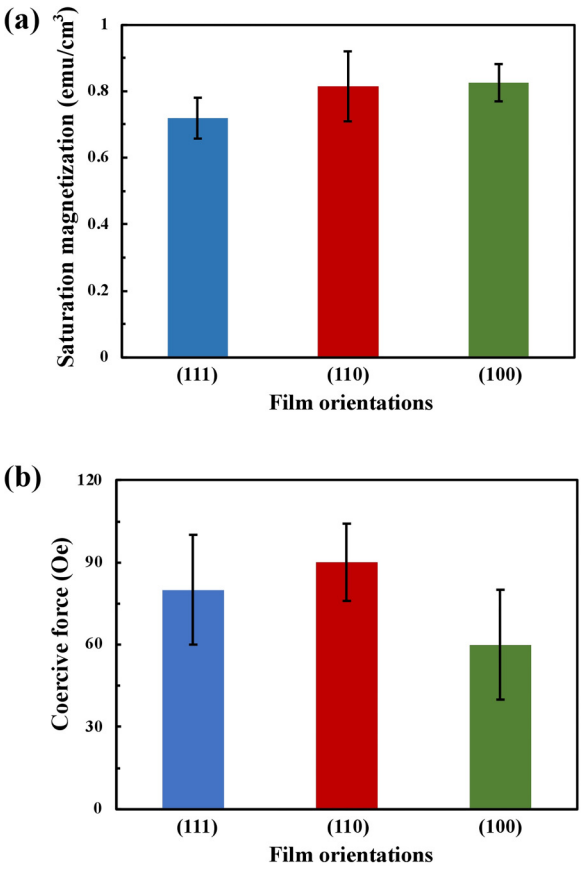


FIG. 9. (a) Saturation magnetization and (b) coercive force of the epitaxial Mn-doped ITO films on YSZ substrates with different crystal plane orientations.

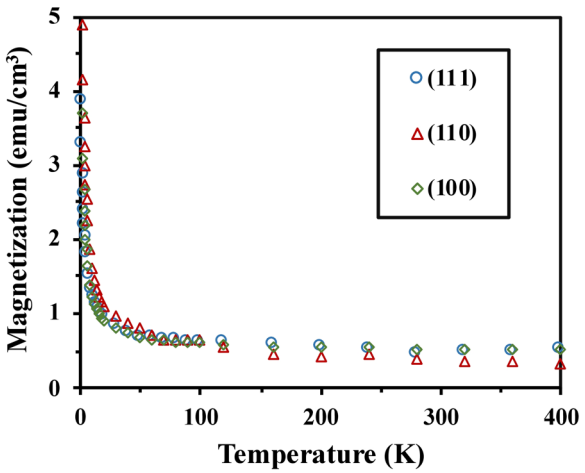


FIG. 10. Temperature dependence of the magnetization of epitaxial Mn-doped ITO films on YSZ substrates with different crystal plane orientations under an applied magnetic field of 1000 Oe.

mechanism of the magnetic properties of the magnetic transparent conducting films.

IV. CONCLUSIONS

In this study, we deposited epitaxial Mn-doped ITO films on YSZ(111), YSZ(110), and YSZ(100) substrates using RF magnetron sputtering. The epitaxial relationship between the Mn-doped ITO films and the YSZ substrates was investigated using XRD measurements. The epitaxial Mn-doped ITO films on the YSZ(111) and YSZ(110) substrates exhibited a high degree of crystalline alignment. All the Mn-doped ITO films exhibited a smooth surface because of epitaxial growth. The Mn-doped ITO film on the YSZ (111) substrate exhibited a particularly low resistivity of $1.34 \times 10^{-4}\,\Omega\,\mathrm{cm}$, high electron mobility of $38.37\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$, and high carrier concentration of $1.21 \times 10^{21}\,\mathrm{cm^{-3}}$. All the Mn-doped ITO films exhibited magnetic hysteresis loops at 300 K. The magnetic properties of the epitaxial Mn-doped ITO films suggest that a combination of delocalized charge carrier-mediated interaction and BMP-driven interaction is required to explain the origin of ferromagnetism in these films. Epitaxial Mn-doped ITO films are one of the most promising candidates for transparent ferromagnetic materials, which have various applications in spintronic devices. Moreover, these films can provide valuable insight into the basic physical properties of magnetic transparent conducting films.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Saiki Kitagawa: Conceptualization (equal); Data curation (lead); Investigation (lead); Validation (lead); Writing – original draft (lead). **Toshihiro Nakamura:** Conceptualization (equal); Funding acquisition (lead); Investigation (supporting); Project

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administration (lead); Supervision (lead); Writing – original draft (supporting); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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